

Curriculum Vitae

Maher Mallem

7 November 1997 (28 y.o.)

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Current position: Postdoctoral researcher at Inria. ROMA team in LIP, ENS Lyon.

Research topics: algorithm design, parameterized complexity theory, scheduling, graph theory.

Web page: <https://perso.ens-lyon.fr/maher.mallem>

Education

Ph.D. student at LIP6, Sorbonne Université

Paris, France

Oct. 2021 - Oct. 2024

- Dates: from 1 October 2021 to 11 October 2024 (three years).
- **Title:** *Parameterized Complexity and New Efficient Enumerative Schemes for RCPSP*.
- Advisor: Claire Hanen (Université Paris Nanterre & LIP6, Sorbonne Université)
- Three-year contract with Sorbonne Université via EDITE.
- Ph.D. defended on 11 October 2024.
 - Jury president: Bruno Escoffier - Professor.
 - Reviewers: Nadia Brauner - Professor, Imed Kacem - Professor.
 - Examiners: Hans L. Bodlaender - Professor, Michael Lampis - Associate Professor.
 - Ph.D. advisor: Claire Hanen - Professor.
- PGMO Ph.D. Award 2025.

Student at École normale supérieure Paris-Saclay

Cachan, France

Sept. 2016 - August 2021

- School diploma obtained in May 2022.
- Master's degree obtained in September 2019. MPRI: Parisian Master of Research in Computer Science.

CPGE student at Lycée Saint-Louis

Paris, France

Sept. 2014 - August 2016

- CPGE are 'higher school preparatory classes' in France.

Experience

LIP, École Normale Supérieure de Lyon

Lyon, France

Nov. 2024 - April 2026

- Postdoctoral contract (project for eighteen months).
- Subject: Scheduling radio-astronomical observations
- Supervisors: Frédéric Vivien - Research Director (Inria), Anne Benoit - Professor (ENS de Lyon).

LIP6, Sorbonne Université

Paris, France

Oct. 2021 - Oct. 2024

- Three-year doctoral contract.
- Subject: Parameterized Complexity and New Efficient Enumerative Schemes for RCPSP
- Advisor : Claire Hanen - Professor (Université Paris Nanterre & LIP6, Sorbonne Université).

(Gap year. Looking for a Ph.D. opportunity.)

Oct. 2020 - Sept. 2021

DIRO, Université de Montréal

Montréal, Canada

Oct. 2019 - July 2020

- ARPE: Full year doing research abroad. Part of my training at ENS Paris-Saclay.
- Subject: Inclusion of horizontal transfers in DL-Super Reconciliation
- Supervisor: Nadia El-Mabrouk - Professor.

LRI, Université Paris-Sud/CNRS

Gif-sur-Yvette, France

March 2019 - August 2019

- M2 internship (four months and a half).
- Subject: RNA constrained design by local search: neighborhood and ergodicity of the sequence space
- Supervisors: Alain Denise - Professor (LRI), Yann Ponty - Research Director (LIX, CNRS/École polytechnique).

DIMAI, Università degli Studi di Firenze

Florence, Italy

March 2018 - July 2018

- M1 internship (four months and a half).
- Subject: Sorting permutations with the DI-machine

- Supervisor: Luca Ferrari - Associate Professor.

LACIM, Université du Québec à Montréal

Montréal, Canada
June 2017 - August 2017

- L3 internship (six weeks).
- Subject: Finding (distinct square factor)-dense infinite words
- Supervisor: Srečko Brlek - Professor.

Teaching

Université Claude Bernard Lyon 1

Lyon, France
Sept. 2025 - Now

- Temporary worker. 46.5 hours planned during the first semester.
Courses: LIFAPSD (algorithms and C++ programming), BDW (databases and web programming).

Polytech Sorbonne

Paris, France
Oct. 2021 - Sept. 2024

- 64 hours a year as part of my Ph.D. contract. Mostly practical work, with a few hours of tutorials occasionally.
- Courses:
 - C language: arrays, pointers, linked lists, structures.
 - Algorithms: sorting algorithms, stacks and queues, hash tables, binary search trees, recursion.
 - Tools for Computer Programming: Git, Linux shell, Makefile.

Year	Course	Level	Number of hours		
			TD	TP	Total
2025-26	Algo., prog. et structures de données	L2	0	24	24
	Base de données et programmation web	L2	0	22.5	22.5
2023-24	Langage C	L3	0	24	24
	Outils pour l'informatique	L3	0	15	15
	Algorithmique élémentaire	L3	0	14	14
2022-23	Informatique 1 : langage C	L3	0	27	27
	Informatique algorithmique	L3	0	16	16
	Informatique langage C	L3	0	16	16
	Algorithmique élémentaire	L3	8	14	22
2021-22	Informatique langage C	L3	0	16	16
	Informatique algorithmique	L3	0	16	16
	Algorithmique élémentaire	L3	14	12	26
-	-	-	-	-	238.5

Research

My research mainly consists in designing efficient algorithms for scheduling problems and studying the computational complexity of these problems.

Motivation

Most scheduling problems are strongly NP-hard and have required intricate heuristics to be solved in practice. While yielding results short term, such heuristics rarely help to identify what fundamentally makes these problems difficult. Such analysis would uncover the similarities between seemingly different problems and facilitate strategy transfers. This would also prove useful whenever one would like to enrich a problem - for example by adding precedence relations or going from equal-length tasks to tasks of arbitrary time length. One way to gather such knowledge is to go beyond classical complexity theory and consider parameterized complexity theory. Given a problem $\mathcal{P}$ we choose parameter k as some property of the input like the number of machines or the width of the precedence graph (if there is one). Depending on problem $\mathcal{P}$ and parameter k either we design algorithms which are *fixed-parameter tractable* with respect to k (i.e. which operate in time $f(k) \times \text{poly}(n)$ with f an arbitrary computable function), or we show that $\mathcal{P}$ remains hard even when k is small (typically via a reduction from a well-known difficult parameterized problem like k -CLIQUE or k -COLORING).

Then the set of parameters for which problem $\mathcal{P}$ is *fixed-parameter tractable* can be interpreted as a footprint. By comparing it to the set of working parameters from another NP-hard problem $\mathcal{P}'$, one could infer whether $\mathcal{P}$ is 'equivalent' to $\mathcal{P}'$, 'strictly harder' than $\mathcal{P}'$, or if both problems are difficult for different reasons.

Current position: postdoctoral researcher at LIP, École Normale Supérieure de Lyon Lyon, France

- Planned duration: eighteen months (i.e., until April 30, 2026). Nov. 2024 - Now
- *Subject : Scheduling radio-astronomical observations*

During this postdoctoral research project supervised by Frédéric Vivien and Anne Benoit, we study the scheduling of radio-astronomical observations. Because the Earth revolves around itself, for a ground-based observer the height of a celestial body in the sky changes over time periodically with a (roughly) twenty-four hour long period. Although one can set a minimum altitude threshold to mitigate the impact of Earth's atmosphere on observation quality, this defines a daily availability window for each celestial body. While scheduling problems featuring collections of availability windows have historically been difficult to optimize and even to approximate, the periodicity of these time windows could play a role in making the problem easier.

This work is done in collaboration with astrophysicists from the ECLAT laboratory, who have shared scheduling data from the NenuFAR radio-telescope over several weeks. During the first few months of this project, we studied the computational complexity - both classic and parameterized - of a non-preemptive model of the problem. This has led to the design of several algorithms - both exact and approximation algorithms - and heuristics. These procedures are currently being implemented and confronted to the NenuFAR data. In the near future, meetings with ECLAT members are planned to discuss the model and find the best compromise between observation quality and makespan minimization.

Previous position: Ph.D. student at LIP6, Sorbonne Université

Paris, France

- Duration : three years. Oct. 2021 - Oct. 2024
- *Subject : Parameterized complexity and new efficient enumerative schemes for RCPSP*

During my Ph.D. thesis supervised by Claire Hanen, I tackled the inclusion of job time windows and precedence delays to RCPSP and its subproblems. With precedence delays we considered parameter ℓ_{max} - i.e. the maximum delay value appearing in the input - in the case of minimum delays, exact delays and the lesser studied maximum delays. For all delay types it was shown that scheduling unit-time jobs on a single machine with a single available delay value is hard even with small ℓ_{max} . In the case of minimum delays we also showed that the parameterized problem remains hard with chains and two distinct available delay values. This suggests that another property of the problem has to be bounded in order to deal with precedence delays.

With job time windows we considered several parameters describing their structure. Notably *slack* σ - i.e. the maximum difference between the time window length of a job and its processing time -, and *pathwidth* μ - i.e. the maximum number of job time windows which can include the same time unit - which have shown recent success in the literature. We showed that scheduling unit-time coupled tasks on a single machine with time windows is para-NP-hard with respect to either parameter.

While this rules out any hope for a *fixed-parameter tractable* algorithm with precedence delays and these parameters alone, the reduction techniques that we used hint at the underlying reason. Indeed unbounded delay values are used to circumvent the job time window restrictions put by these parameters. And fittingly we showed that the problem becomes *fixed-parameter tractable* if you combine them with maximum delay value ℓ_{max} . In fact we can even have general precedence with any number of distinct delay values, either on parallel machines or with arbitrary processing times. With multiple machines and arbitrary processing times the problem becomes hard again, and you would have to add an extra parameter - like the maximum processing time - to get a *fixed-parameter tractable* algorithm.

Finally with job time windows and no precedence delays we introduced a new parameter called the *proper level* pl . Parameter pl is defined as the maximum number of job time windows which can strictly include - on both ends - another job time window. On single machine scheduling with job time windows we proposed a new dominance rule which generalizes the *Earliest Deadline* rule and leads to a *fixed-parameter tractable* number of dominant schedules with respect to pl . The resulting makespan algorithm is quasi-linear in the input size with a single exponential dependency on pl , which gives it a good chance to be competitive in future applications.

For this work, I received the PGMO Ph.D. Award 2025.

Publications

[Google Scholar page](#). [DBLP page](#). All preprints are available [in this page](#).

International journals

- *Journal of Combinatorial Optimization* March 2026
 - **Title:** *Single Machine Scheduling with Precedence Constraints and Bounded Maximum Delay Value*
 - Authors: Maher Mallem, Claire Hanen and Alix Munier-Kordon (LIP6).
 - Regular paper (32 pages). [DOI](#).
- *Theoretical Computer Science* Feb. 2026
 - **Title:** *Coupled-task Scheduling with Time Windows, Bounded Pathwidth and Bounded Slack is para-NP-complete*
 - Authors: Maher Mallem, Claire Hanen and Alix Munier-Kordon (LIP6).
 - Regular paper (20 pages). [DOI](#).

International conferences with program committees and proceedings

- [IWOCA 2026](#) Clermont-Ferrand, France
 - **Title:** *Parameterized Complexity Results for Scheduling with Precedence Delays: Shuffle Product and Directed Bandwidth* June 2026 (accepted)
 - Authors: Hans L. Bodlaender (Utrecht University) and Maher Mallem.
 - Regular paper (25 pages).
- [FUN 2026](#) Porquerolles, France
 - Full name: International Conference on Fun with Algorithms May 2026 (accepted)
 - **Title:** *Finding Shortest Paths in Kuru Kuru Kururin*
 - Authors: Mickaël Laurent (Charles University) and Maher Mallem.
 - Regular paper (19 pages).
- [ISCO 2024](#) Tenerife, Canary Islands, Spain
 - Full name: International Symposium on Combinatorial Optimization May 2024
 - **Title:** *A New Structural Parameter on Single Machine Scheduling with Release Dates and Deadlines*
 - Authors: Maher Mallem, Claire Hanen and Alix Munier-Kordon (LIP6).
 - Regular paper (15 pages). [DOI](#).
- [IPEC 2022](#) Potsdam, Germany
 - Full name: International Symposium on Parameterized and Exact Computation Sept. 2022
 - **Title:** *Parameterized Complexity of a Parallel Machine Scheduling Problem*
 - Authors: Maher Mallem, Claire Hanen and Alix Munier-Kordon (LIP6).
 - Regular paper (21 pages). [DOI](#).

Other international conferences

- [PMS 2026](#) Toulouse, France
 - Full name: International Workshop on Project Management and Scheduling April 2026 (accepted)
 - **Title:** *Notes on Scheduling Radio-astronomical Observations*
 - Authors: Maher Mallem, Frédéric Vivien (LIP).
 - Extended abstract (4 pages).
- [MAPSP 2022](#) Oropa (Biella), Italy
 - Full name: Workshop on Models and Algorithms for Planning and Scheduling June 2022
 - **Title:** *Parameterized Complexity of a Single Machine Scheduling Problem with Precedence, Release Dates and Deadlines*
 - Authors: Claire Hanen, Maher Mallem and Alix Munier-Kordon (LIP6).
 - Extended abstract (3 pages).

National conferences

- ROADEF 2026 Tours, France
Feb. 2026
 - Full name: Congrès annuel de la Société Française de Recherche Opérationnelle et d'Aide à la Décision
 - **Title:** *Computational Complexity of Radio-astronomical Observation Scheduling*
 - Authors: Maher Mallem, Frédéric Vivien (LIP).
 - Abstract (2 pages).
- ROADEF 2024 Amiens, France
March 2024
 - Full name: Congrès annuel de la Société Française de Recherche Opérationnelle et d'Aide à la Décision
 - **Title:** *Parameterized Complexity: a Two-Dimensional Approach to Study Scheduling Problems*
 - Authors: Maher Mallem, Claire Hanen and Alix Munier-Kordon (LIP6).
 - Abstract (2 pages).
- ROADEF 2022 Lyon, France
Feb. 2022
 - Full name: Congrès annuel de la Société Française de Recherche Opérationnelle et d'Aide à la Décision
 - **Title:** *Parameterized Complexity of a Single Machine Scheduling Problem*
 - Author: Maher Mallem.
 - Abstract (2 pages).
- SeqBIM 2019 Marne-la-Vallée, France
Dec. 2019
 - Theme: Sequences in Bioinformatics, Computer Science and Mathematics
 - **Title:** *Forbidden Substrings and the Connectivity of the Hamming Graph of RNA Sequences: Partial Disconnectivity Tests*
 - Authors: Maher Mallem, Alain Denise (LRI) and Yann Ponty (LIX, CNRS/École polytechnique).
 - Extended abstract (6 pages).

Submitted papers

- *INFORMS Journal on Computing* Submitted in Nov. 2024
(under minor revision)
 - **Title:** *Kernelization Algorithms on Single Machine Scheduling with Time Windows*
 - Authors: Maher Mallem, Claire Hanen and Alix Munier-Kordon (LIP6).

In preparation

- *Discrete Applied Mathematics* Writing in progress
 - **Title:** *Scheduling with Precedence Constraints, Time Windows and Bounded Proper Level*
 - Authors: Maher Mallem, Claire Hanen and Alix Munier-Kordon (LIP6).

Workshop presentations

- MoMA 2025 school-workshop Aussois, France
Nov. 2025
 - Full name: Modern Optimization Methods and Models in Algorithm Design.
 - **Title:** *Scheduling astronomical observations*
- ROMA team seminar (LIP) Lyon, France
Nov. 2025
 - **Title:** *Introduction to fixed-parameter tractability: Single machine scheduling with release dates and deadlines*
- Temporary member seminar series (LIP) Lyon, France
Nov. 2025
 - **Title:** *An introduction to scheduling: The case of astronomical observations*
- PGMO 2025 Ph.D. prize ceremony Palaiseau, France
Nov. 2025
 - **Title:** *Parameterized complexity and new efficient enumerative schemes for RCPSP*
- “Algorithms and Complexity” group seminar at Utrecht University Utrecht, The Netherlands
Nov. 2025
 - **Title:** *Scheduling with release dates and deadlines: going beyond the pathwidth*
- VERIMAG laboratory seminar Grenoble, France
May 2025
 - **Title:** *Parameterized complexity of scheduling problems with precedence delays*

- **MC2 team seminar (LIP)** Lyon, France
 - *Title: Parameterized complexity of single machine scheduling with job time windows* March 2025
- **ENS Lyon 2025 sports-study week** Le Pleyne, France
 - Ninety-minute long presentation to the ENS Lyon first year computer science students. Jan. 2025
 - *Title: Scheduling and parameterized complexity*
- **CoA 2024 workshop** Paris, France
 - Full name: Complexity and Algorithms. Nov. 2024
 - *Title: Kernelization algorithms on machine scheduling with time windows*
- **RO team seminar (LIP6)** Paris, France
 - *Title: A new structural parameter on single machine scheduling with release dates and deadlines* May 2024
- **Masterclass of the Sorbonne Université Colloquium** Paris, France
 - Invited researcher: Claire Mathieu (IRIF, CNRS/Université de Paris). Jan. 2024
 - *Title: Complexité paramétrée des problèmes 1-Machine avec fenêtres temporelles*
- **Complexity Days workshop** Paris, France
 - Organized at Université Paris Cité by GdR IM. Dec. 2023
 - *Title: Scheduling meets parameterized complexity*
- **AGaPe/GOTHA working group days** Paris, France
 - Organized at Sorbonne Université by GdR ROD. Nov. 2023
 - Theme: Scheduling and algorithms with performance guarantee.
 - *Title: Scheduling meets parameterized complexity*
- **OPA/REST working group days** Tours, France
 - Organized at Polytech Tours by GdR ROD. Oct. 2023
 - *Title: Parameterized complexity of parallel machine scheduling with release dates, deadlines and precedence relations*

Skills & Interests

Programming: Experienced: C, Python. Intermediate: OCaml, C++. Basic: Gallina (Rocq), Javascript, Lua.

Language: French (mother), English (C1 - almost bilingual).

Interests:

- Tool-Assisted Speedrunning 2019 - Now
- Speedrunning 2016 - Now
- Recreational mathematics 2014 - Now